5 Stability Analysis of Control Systems

CONCEPT OF STABILITY

- A system is stable if the system eventually comes back to the equilibrium state when the system is subjected to an initial condition.
- Stability is the most desired property in the design of control systems.
- A system is unstable if the output diverges without bound from its equilibrium state when the system is subjected to an initial condition.
- A system is marginally stable if the system tends to oscillate about its equilibrium state when the system is subjected to an initial condition.

Rack your Brain

Why stability is important in 'control system'?

Stability of LTI system:

- An LTI system is stable if it produces bounded output for every bounded input.
- The stability of a system depends upon poles.
- If all the poles are located in the left hand side of the s-plane, then the system is stable.
- A system is said to be asymptotically stable, if in the absence of any input, the output tends to zero regardless of the initial condition.

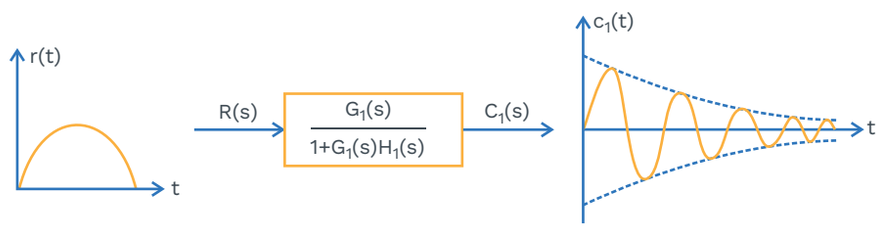
Bounded signal

The signal which is finite at every point of time is said to be a bounded signal.

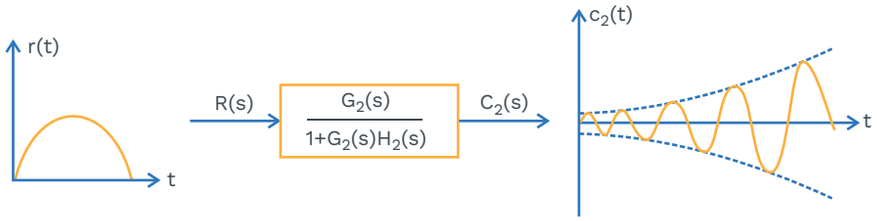
BIBO Stability Criterion:

- A system is stable if for every bounded input signal the system response is bounded.
- A system is unstable if for any bounded input signal the system response is unbounded.

For example: An illustration to explain stability and instability is shown below–



(a) Stable System



(b) Unstable System

Figure-5.1

Types of stability for a system:

The stability can easily be classified as follows for analysis and design:

1. Absolute stability
2. Relative stability

Absolute stability:

- A system that provides bounded output despite variation in its parameters is considered absolutely stable.
- The term absolute stability is used in the control system for qualitative analysis of stability.
- Absolute stability determines the stable or unstable condition of the system.
- In this type of system, the output after reaching steady-state doesn't show changes regardless of the variations in the system parameter values.

The figure 6.2 illustrates the step response of an absolutely stable system.

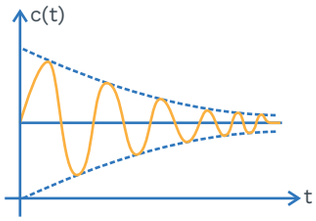


Figure-5.2 Response of absolutely stable system

- Poles must have real and negative nature for an absolutely stable condition to exist.
- The method of determining absolute stability is given by Routh-Hurwitz Criterion.

Relative stability:

- Relative stability is defined as a quantitative measure of the transients die out time in the system.
- Relative stability is calculated by using the relative settling time of each pair of roots and settling time is inversely related to the real part of the roots.
- The relative stability is determined by using frequency response methods.
- If a root (or a pair of roots) of a characteristic equation moves further away from the imaginary axis. The relative stability of the system improves, which can be shown as–

Rack your Brain

State the difference between relative stability and absolute stability.

Gray Matter Alert!!!

From the position of the roots of the characteristic equation in the s-plane, we can determine the stability of the system.

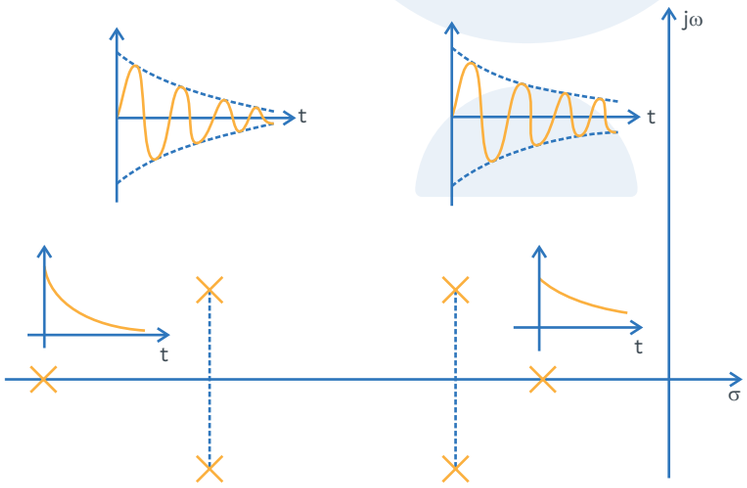


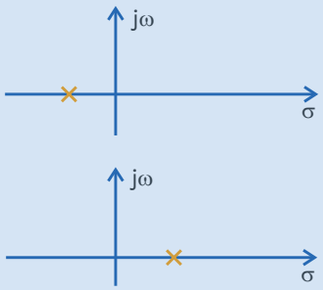
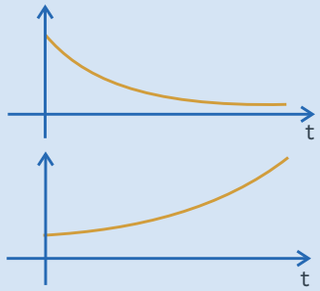
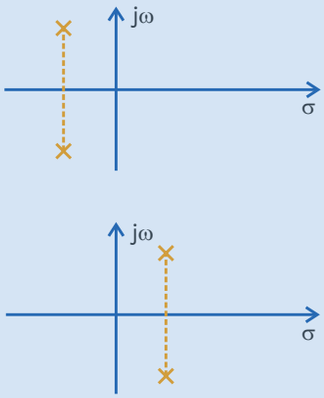
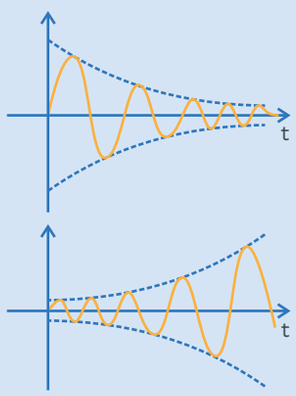
Figure-5.3 Relative stability for various root location in s-plane

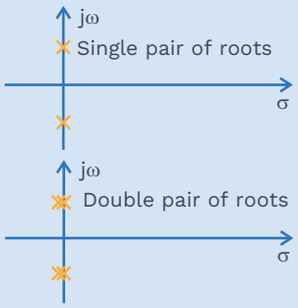
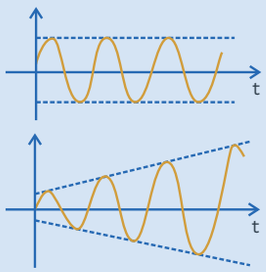
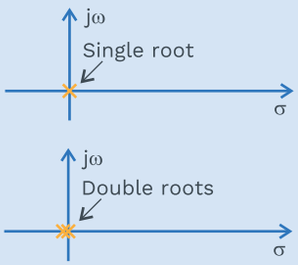
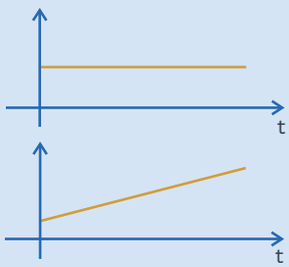
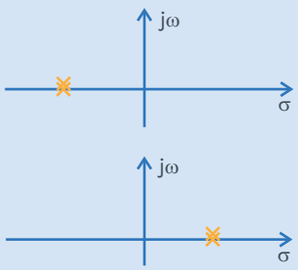
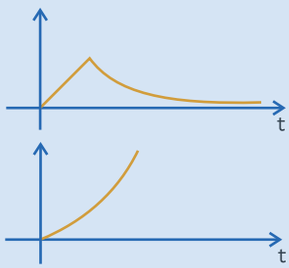
- Relative stability is measured in terms of gain margin, phase margin, maximum overshoot and damping ratio.

Stability relationship with poles:

- The impulse response for a characteristic equation having all the roots negative and real parts remains bounded and thus eventually reduces to zero.

- The roots of a characteristic equation with positive real parts have an unbounded impulse response and are termed BIBO unstable systems.
- If the characteristic equation has repeated roots on the imaginary axis then the response is infinite and unbounded, the system is BIBO unstable system.
- The first condition of stability satisfies only if the impulse response is absolutely integrable, and the nature of impulse response depends on the poles of transfer function G(s), which are the roots of the characteristic equation. Roots may be both real and complex conjugate, and a multiplicity of any order may exist. The nature of the response of all possible types of roots are shown in the table below.

	Roots in s-plane	Corresponding impulse Response	Stability Status
1.			Absolutely stable Unstable
2.			Absolutely stable Unstable

	Roots in s-plane	Corresponding impulse Response	Stability Status
3.			Marginally stable Unstable
4.			Marginally stable Unstable
5.			Absolutely stable Unstable

- One or more non-repeated roots of the characteristic equation must lie on the imaginary axis for an impulse response to be said to be bounded.
- In the absence of any common pole, the output is bounded and has sustained oscillations. These kinds of systems are called 'marginally stable.

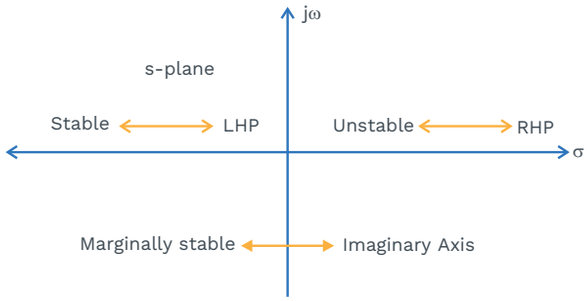


Figure-5.4 Stability Relationship with poles

Example-1.
If the non-reapeted poles of the system exists on the jω axis then this system is said to be-
(a) Stable
(b) Unstable
(c) Conditionally stable
(d) Marginally stable

Solution:
If the system has one or more than one non repeated roots lying on the imaginary axis then it is said to be a marginally stable system.
Hence, the correct option is (d).

Previous Year's Questions

None of the poles of a linear control system lie in the right half of s-plane for a bounded input, the output of this system is–
(a) is always bounded
(b) could be unbounded
(c) always tends to zero
(d) none of the above
Answer: (d)

Condition for stability:

- Consider a system with general form of transfer function,

$$T(s) = \frac{p(s)}{q(s)} = \frac{b_0 s^m + b_1 s^{m-1} + \ldots\ldots\ldots b_{m-1} s + b_m}{a_0 s^n + a_1 s^{n-1} + \ldots\ldots\ldots a_{n-1} s + a_n}$$

- The characteristic equation of the system is given by

$$q(s) = a_0 s^n + a_1 s^{n-1} + \ldots\ldots\ldots a_{n-1} s + a_n = 0$$

In the above equation, we assume that $a_0 > 0$
For case $a_0 < 0$, it can be changed to a positive value by multiplying the characteristic equation throughout by a factor of -1.

- For stability, it is necessary to determine whether any roots of the characteristic equation lies in the right half of plane (RHP) of the s–plane.

1. The essential condition for the stability of the first and second order system is the positive nature of coefficients of the characteristic equation. The characteristic equation of a first order system is,

$a_0 s + a_1 = 0$ which has a single root.

Root $s_1 = -\frac{a_1}{a_0}$

The positiveness of a_0 and a_1 ensures a negative root which implies the stability of the system.

2. The characteristic equation of a second order system is

$a_0 s^2 + a_1 s + a_2 = 0$ which has two roots.

Roots $s_1, s_2 = \frac{\left[-a_1 \pm \sqrt{a_1^2 - 4a_0 a_2}\right]}{2a_0}$

The positiveness of the a_0, a_1 and a_2 ensures that the roots lie in the left half of the s-plane which implies the stability of the system.

3. But these conditions are not sufficient to ensure the stability of third and higher order systems.
4. Consider a third order system and it's characteristic equation is given by -

$$s^3 + s^2 + 2s + 8 = 0$$

The roots of the equation are,

$$s = -2,\ 0.5 \pm j\ 1.936$$

Indicating the system to be unstable while all the coefficients are positive.

Example-2.

Select the appropriate transfer function from the following which are minimum phase transfer function.

(1) $\dfrac{1}{(s-1)}$ (2) $\dfrac{(s-1)}{(s+3)(s+4)}$

(3) $\dfrac{(s+2)}{(s+3)(s+4)}$

Choose the correct option from the given codes:
(a) Both (2) and (3)
(b) (2) only
(c) Both (1) and (3)
(d) (3) only

Solution:

For the minimum phase transfer function, zeros or poles should not lie on the right side of the s-plane.
Hence, the correct option is (d).

Example-3.

The characteristic equation is given by $s^3 - 2s + 2 = 0$.
The number of positive real roots of the given equation is -

Solution:

$$s^3 - 2s + 2 = 0$$
$$s = -1.77,\ 0.88 \pm j\ 0.59$$

Hence, no positive real roots.

Gray Matter Alert!!!

- The characteristic equation should be first verified for the analysis of the stability of the system.
- When some coefficients are negative or zero, the system is considered unstable.
- If all the coefficients are of the same sign i.e., either positive or negative then there is a possibility of stability then one should move further to test the sufficient condition of stability.

ROUTH–HURWITZ CRITERION

- Routh Hurwitz's stability criterion provides a sufficient and necessary condition to analyze the system's stability
- "Routh Hurwitz criterion states that the number of roots of the characteristic equation with positive real parts is equivalent to the number of changes in sign of the first column of the Routh array."
- The characteristic equation of an nth order system is

$$q(s) = a_0 s^n + a_1 s^{n-1} + \ldots\ldots\ldots\ldots a_{n-1} s + a_n = 0$$

For the stability of this system, it is necessary and sufficient that the determinants formed from the coefficients a_0, a_1 ..... a_n of the characteristic equation be positive.

- Routh array is defined as an array formed by the ordering of the coefficients of a characteristic equation based on Routh Hurwitz criterion which is given as –

s^n	a_0	a_2	a_4	a_6	.	.
s^{n-1}	a_1	a_3	a_5	.	.	.
s^{n-2}	b_1	b_2	b_3	.	.	.
s^{n-3}	c_1	c_2				
s^{n-4}	d_1	d_2				
.	.	.				
.	.	.				
.	.	.				
.						
s^0	a_n					

The coefficients b_1, b_2 ..... are evaluated as –

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}$$

$$b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

Similarly, $$c_1 = \frac{b_1 a_3 - a_1 b_2}{b_1}$$

$$c_2 = \frac{b_1 a_5 - a_1 b_3}{b_1}$$

The continuous process takes place until a zero is obtained as a last coefficient in the third row.

The Routh stability criterion states that if the first column of the Routh array does not experience any sign change i.e. all the terms are positive then the given system is stable, else the system is unstable. If the system is unstable then the number of sign changes in the first column of the Routh array is equal to the number of roots that lies on the right half of the s-plane.

Conclusions

1. If any term is negative, then there is atleast one root on the right half of the imaginary axis and the system is unstable.
2. The number of sign changes in the first column of the Routh array is equal to the number of roots located on the right side of an imaginary axis.

Example–4.

The characteristic equation for a system is given by –

$$s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$$

Determine the stability using Routh-Hurwitz Criterion.

Solution:

The characteristic equation is –

$$s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$$

The Routh array for the given characteristic equation is –

s^4	1	18	5
s^3	8	16	0
s^2	16	5	
s^1	13.5	0	
s^0	5		

The elements of the first column are all positive. Hence, the system is stable.

Previous Year's Questions

Consider the following statements

(1) A system is said to be stable if its output is bounded for any input.

(2) A system is stable if all the roots of the characteristic equation lie in the left half of s-plane.

(3) A system is stable if all the roots of the characteristic equation have negative real parts.

(4) A second order system is always stable for finite positive values of open loop gain.

Which of the above statements are correct–

(a) (2), (3) and (4)

(b) (1) only

(c) (2) and (3) only

(d) (3) and (4) only

Answer: (c)

Example-5.
A negative feedback control system has an open-loop transfer function having two poles, two zeros and a variable gain K. Compute the range of K using the Routh Hurwitz criterion for which the closed-loop system has 0, 1 or 2 poles in the right half of the s-plane.

$$G(s) = \frac{K(s+1)(s+2)}{(s+0.1)(s-1)}$$

Solution:

Given that, $G(s) = \frac{K(s+1)(s+2)}{(s+0.1)(s-1)}$

Characteristic equation -

$$1 + G(s) = 0$$

$$1 + \frac{K(s+1)(s+2)}{(s+0.1)(s-1)} = 0$$

$$\therefore \quad (1 + K)s^2 + (3K - 0.9)s + (2K - 0.1) = 0$$

Routh array is –

s^2	1 + K	2K – 0.1
s^1	3K – 0.9	0
s^0	2K – 0.1	

Rack your Brain

What is the application of Routh Criterion?

(i) No poles in the right half of the s-plane

K + 1 > 0	or	K > –1
3K – 0.9 > 0	or	K > 0.3
2K – 0.1 > 0	or	K > 0.05

Hence, K > 0.3

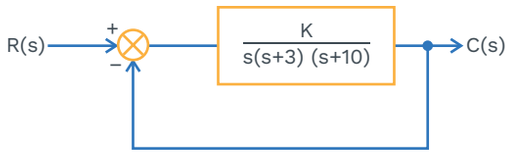
(ii) One pole in the right half of the s-plane = one sign change in the first column.

$$-1 < K < 0.05$$

(iii) Two poles in the right half of the s-plane = two sign changes in the first column.

$$0.05 < K < 0.3$$

Example-6.
A feedback system is shown in the given figure, here K > 0.



Compute the range of K for which the system attains stability.

Solution:

Given, $G(s) = \dfrac{K}{s(s+3)(s+10)}$

And $H(s) = 1$

The characteristic equation is –

$$1 + G(s)\,H(s) = 0$$

$$1 + \frac{K}{s(s+3)(s+10)} = 0$$

$$s(s + 3)\,(s + 10) + K = 0$$

$$s^3 + 13s^2 + 30s + K = 0$$

Routh array is –

s^3	1	30
s^2	13	K
s^1	$\dfrac{13 \times 30 - K}{13}$	0
s^0	K	

For a stable system no sign changes are observed in the first column of the Routh array.

$\therefore$ K > 0

And $\dfrac{13 \times 30 - K}{13} > 0$

$\therefore$ The range of K for a stable system is 0 < K < 390.

Example-7.
The open loop transfer function is given below.

$$G(s) = \frac{e^{-sT}}{s(s+2)}$$

Compute the stability condition of the system with the help of Routh Hurwitz criterion.

Solution:

Overall transfer function is

$$\frac{C(s)}{R(s)} = \frac{\frac{e^{-sT}}{s(s+2)}}{1+\frac{e^{-sT}}{s(s+2)}.1}$$

$$\frac{C(s)}{R(s)} = \frac{e^{-sT}}{s^2+2s+e^{-sT}}$$

The characteristic equation is –

$$s^2 + 2s + e^{-sT} = 0$$

Since $e^{-sT} = 1 - sT + \frac{s^2T^2}{2!} + ...$

Considering only first two terms of series, characteristic equation can be written as–

$$s^2 + 2s + (1 - sT) = 0$$

$$s^2 + (2 - T)s + 1 = 0$$

Routh array is –

s^2	1	1
s^1	2 – T	0
s^0	$\frac{(2-T)-0}{2-T}$	0

Condition for stability is given by –

$$(2 - T) > 0$$

$$T < 2$$

Special case:

The coefficients of the equation may cause the following difficulties:

1. An element of the first column of the Routh array is zero.
2. In the Routh array, if all the elements of a row are equal to zero.

Case-1. Zero in the first column

- Whenever an element in a row becomes zero in a Routh array, the next row becomes infinite, and the Routh array is stopped from being computed.
- In this case, zero is replaced by a small positive number (say ε) and compute the rest of the Routh array using ε.

Previous Year's Questions

The number of roots of the equation,

$$2s^4 + s^3 + 3s^2 + 5s + 7 = 0$$

That lie in the right half of s-plane is

(a) Zero (b) One
(c) Two (d) Three

Answer: (b)

- Lastly, let $\varepsilon \to 0$; to test the sign of the first column of the array. The following examples illustrate these methods.

Example–8.
Evaluate the stability of a closed loop system whose characteristic equation is,

$$s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$$

Solution:

$$s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$$

Routh array is -

s^5	1	2	3
s^4	1	2	5
s^3	0	–2	

The first element of s^3 row is zero. Hence, the element in the s^2 row is infinite. To overcome this, we replace zero in the s^3 row by a small positive number (ε) and then proceed with the computation of the array.

The Routh table is modified as –

s^5	1	2	3
s^4	1	2	5
s^3	ε	–2	0
s^2	$\lim_{\varepsilon\to 0}\left(\frac{2\varepsilon+2}{\varepsilon}\right) \to +\infty$	5	0
s^1	$\lim_{\varepsilon\to 0}\left(\frac{-4\varepsilon-4-5\varepsilon^2}{2\varepsilon+2}\right) \to -2$	0	0
s^0	5		

The first column shows two signs, which indicates that two poles of the system exists on the right half of the s-plane which makes the system unstable.

Example–9.
Evaluate the stability of a system whose characteristic equation is given as,

$$s^5 + 2s^4 + 3s^3 + 6s^2 + 2s + 1 = 0$$

Also, calculate the number of roots that lies on the right side of the s-plane.

Solution:
Characteristic equation is–

$$s^5 + 2s^4 + 3s^3 + 6s^2 + 2s + 1 = 0$$

Routh table as–

	s^5	1	3	2
	s^4	2	6	1
	s^3	ε	1.5	
Sign change –	s^2	$\lim_{\varepsilon \to 0}\left(\frac{6\varepsilon - 3}{\varepsilon}\right) \to -\infty$	1	
Sign change +	s^1	$\lim_{\varepsilon \to 0}\left(\frac{9\varepsilon - 4.5 - \varepsilon^2}{6\varepsilon - 3}\right) \to +1.5$	0	
	s^0	1		

Here two sign changes in the first column.

$\therefore$ System is unstable.

Number of sign changes = 2

$\therefore$ Number of poles that lies in RHS = Number of sign change = 2

Example-10.

Consider the following characteristic equation

$$(s + 1)(s + 2)(s^2 + 1) = 0$$

Determine its stability state.

Solution:

$$(s + 1)(s + 2)(s^2 + 1) = 0$$

$$s^4 + 3s^3 + 3s^2 + 3s + 2 = 0$$

Routh table is as follows–

s^4	1	3	2
s^3	3	3	
s^2	2	2	
s^1	0 (Here we will insert real positive number 'ε')		
s^0	2		

The elements of the first column are all positive. Hence, the system is stable

Rack your Brain

What is the necessary condition for which the system is stated as a stable system?

Alternate method for the case-1

To overcome the difficulties caused by a zero term in the first column of the Routh array, we can use the second method also.

- Modify the original characteristic equation by replacing s by z^{-1}.
- Apply the Routh criterion on the modified equation in terms of z.

- The number of roots in the z-plane having positive real parts is the same as the number of roots in the s-plane having positive real parts.

Consider the characteristic equation is –

$$s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$$

Substitute s by z^{-1} in the characteristic equation and rearrange the terms to obtain an equation in terms of z.

$$5z^5 + 3z^4 + 2z^3 + 2z^2 + z + 1 = 0$$

The Routh array for this equation is–

z^5	5	2	1
z^4	3	2	1
z^3	–4/3	–2/3	0
z^2	1/2	1	0
z^1	2	0	
z^0	1		

There are two sign changes in the first column of the Routh array which means that there are two roots outside the unit circle of the z-plane. Hence, there are two roots in the right half of the s-plane.

Case-2. When entire row of the Routh array is zero–

- Consider a system with characteristic equation

$$s^3 + 5s^2 + 6s + 30 = 0 \qquad \text{.....(i)}$$

- Routh table of equation as–

s^3	1	6
s^2	5	30
s^1	0	0

- An entire row of zero indicates the symmetrically located roots in the s-plane and real roots with opposite sign forming a pair or pair of conjugate roots on the imaginary axis or complex conjugate roots in the s-plane.
- If an entire row is zero, an auxiliary polynomial is formed with the entry of the row immediately above.
- The auxiliary polynomial is differentiated w.r.t 's'.
- The derivative of auxiliary equation gives the coefficients to replace row of zeros and the Routh table is interpreted along with the roots of the auxiliary polynomial.
- From the Routh table of equation (i), interprets the change in sign of coefficients in the first column as:

s^3	1	6
s^2	5	30
s^1	0	0

The auxiliary equation is given by,

$$A(s) = 5s^2 + 30$$

$$5s^2 + 30 = 0$$

$\frac{dA(s)}{ds} = 10s + 0$; it governs s^1 row

Thus in the Routh table, the third row is replaced with coefficients 10 and 0.

s^3	1	6
s^2	5	30
s^1	10	0
s^0	30	

- All the elements in the first column are positive. Hence, the system is marginally stable as the roots of the auxiliary equation lie on the imaginary axis.

Example-11.

The characteristic equation of sixth order system is given as–

$$s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$$

Determine the stability of the system.

Solution–

Given characteristic equation is,

$$s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$$

Now plotting Routh array table,

s^6	1	8	20	16
s^5	2	12	16	
s^4	2	12	16	
s^3	0	0	0 ← Entire row is zero, so we will find auxiliary equation	
s^2				
s				

Now, the auxiliary equation is given by

$$A(s) = 2s^4 + 12s^2 + 16$$

By taking differentiation w.r.t 's' we get -

$$\frac{dA(s)}{ds} = 8s^3 + 24s \qquad \text{.....(i)}$$

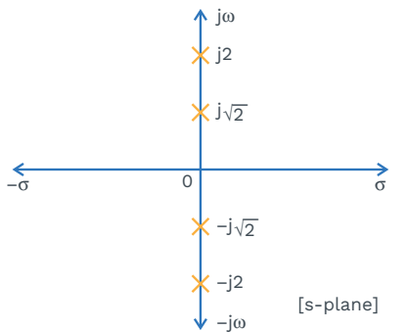
Now, we will put the value of equation (i) in 3rd row.

s^6	1	8	20	16
s^5	2	12	16	
s^4	2	12	16	
s^3	8	24		
s^2	6	16		
s^1	2.66			
s^0	16			

There is no sign change observed in the first column of the Routh table.
Now, for the root, we will solve the auxiliary equation,

$\therefore$ $s^4 + 6s^2 + 8 = 0$
Let $s^2 = x$
$\therefore$ $x^2 + 6x + 8 = 0$
$\Rightarrow$ $x^2 + 4x + 2x + 8 = 0$
$\Rightarrow$ $x(x + 4) + 2(x + 4) = 0$
$(x + 4)(x + 2) = 0$
$\therefore$ $x = -2$; $x = -4$
$s^2 = -2$; $s^2 = -4$
$s = \pm j\sqrt{2}$; $s = \pm j2$

Now locating the above poles on s-plane.



[s-plane]

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The order of auxiliary polynomial is always even.

The roots above are located on the jω axis, so the system is marginally stable, and the 1st column does not show any sign changes, so other two roots lies on the left half of the s-plane.

Properties of the auxiliary equation –

1. The roots of the auxiliary equation are also roots of the characteristic equation.
2. The roots of the auxiliary equation are symmetrical about an imaginary axis.
3. If an auxiliary equation has an odd order then it has a pole at the origin.

Some important concepts –

1. In the Routh table when the row of zeros occurs and all the coefficients in the first column have the same sign then the system is marginally stable because the non repeated poles must lie on the imaginary axis.
2. If the number of times odd row of zero occurs, it indicates that there will be repeated symmetrical poles w.r.t the origin.
3. If the sign change occurs above the auxiliary equation then there will be no symmetrical pole in LHS of the s-plane corresponding to the pole which is in the right half of the s-plane.
4. If the sign change occurs below the auxiliary equation then there must be a symmetrical pole in LHS of the s-plane corresponding to the pole placed in the RHS of the s-plane.
5. When many times row of zero occurs and all the coefficients of the first column are positive then the system is unstable because the poles that will lie on imaginary axis will be repeated.

Example-12.

The open loop transfer function of a system is $G(s) = \dfrac{K}{s(s+2)(s^2+2s+2)}$ is connected to a negative feedback configuration having unity feedback gain. What will be the value of K for which the closed-loop system is marginally stable?

Solution:

Given, $G(s) = \dfrac{K}{s(s+2)(s^2+2s+2)}$

The characteristic equation of negative feedback control system is given by,

$$1 + G(s)\,H(s) = 0$$

$$1 + \frac{K}{s(s+2)(s^2+2s+2)} = 0$$

$$s^4 + 4s^3 + 6s^2 + 4s + K = 0$$

Routh table is as follow –

s^4	1	6	K
s^3	4	4	0
s^2	5	K	0
s^1	$\dfrac{20-4K}{5}$	0	0
s^0	K	0	0

For system to be marginally stable odd row of Routh Hurwitz array must be zero and there is no sign change in first column,

$$K > 0$$

And $$\frac{20 - 4K}{5} = 0$$

$$K = 5$$

Hence, for the given system to be marginally stable,

$$K = 5$$

Example-13.

The first element of each row of a Routh Hurwitz stability test is shown below with their sign.

Row	I	II	III	IV	V
Sign	−	+	+	+	−

The considered statements are as follows:

(1) The system has two roots in 1st and 4th quadrant of the s-plane
(2) System is stable
(3) System is unstable
(4) There is a total of 2 repeated poles

(a) (1) and (4) (b) (1) and (2)
(c) (1), (2) and (4) (d) (1) and (3)

Solution:

Total number of changes of sign = 2
∴ Number of roots on RHS = 2
∴ System is unstable
Hence, the correct option (d).

Example-14.

Determine the frequency of sustained oscillation, if the loop gain K of a negative feedback system with open-loop transfer function is given as,

$$G(s)\,H(s) = \frac{K(s+3)}{(s+8)^2}.$$

Solution:

$$\text{OLTF, } G(s)\,H(s) = \frac{K(s+3)}{(s+8)^2}$$

∴ The characteristic equation is -

$$1 + G(s)\,H(s) = 0$$

$$1 + \frac{K(s+3)}{(s+8)^2} = 0$$

$$s^2 + (16 + K)s + (64 + 3K) = 0$$

Routh array is -

s^2	1	64 + 3K
s^1	16 + K	
s^0	64 + 3K	

For sustained oscillation,

$$16 + K = 0$$

$$K = -16$$

Auxiliary equation is -

$$s^2 + 64 + 3K = 0$$

$$s^2 + 64 + 3(-16) = 0$$

$$s^2 + 16 = 0$$

$$s = \pm j\,4$$

Frequency of oscillation = 4 rad/sec

Example-15.

The characteristic equation of a closed loop system is given by -

$1 + G(s)\,H(s) = s^4 + s^3 - 3s^2 - s + 2$

Calculate all the poles.

Solution:

Given, characteristic equation is

$$s^4 + s^3 - 3s^2 - s + 2 = 0$$

Routh table is -

s^4	1	−3	2
s^3	1	−1	
s^2	−2	2	
s^1	0	0←Entire row is zero, so calculate auxiliary equation	
s^0			

$$A(s) = -2s^2 + 2$$

$$\frac{dA(s)}{ds} = -4s \qquad \text{.....(i)}$$

Now, putting the value of equation (i) in row 4, we get -

s^4 1 –3 2

s^3 1 –1

s^2 –2 2

s^1 –4

s^0 2

For the calculation of poles, we have to find the roots of auxiliary equation.

$\therefore \quad -2s^2 + 2 = 0$

$\Rightarrow \quad s^2 = 1 \qquad \Rightarrow s = \pm 1$

As there is only one sign change above zero line, so one of the poles lie on the right half of the s-plane.

From the auxiliary equation, one root is s = 1, lies on the right side of the s-plane.

So, the other two poles lie on the left of the s-plane.

Relative stability analysis –

- In order to determine the relative stability of a system, we calculate the settling time of the dominant roots of its characteristic equation.
- Settling time has an inverse relationship with the real part of dominant roots.
- When all the roots of the characteristic equation are more negative than a specific value, the relative stability may be specified.
- All the roots must lie to the left of the lines $s = -\sigma_1$ ($\sigma_1 > 0$)
- The characteristic equation of the system is modified further by shifting the origin of the s-plane to $s = -\sigma_1$ i.e. by substitution $s = z - \sigma_1$ as shown below –

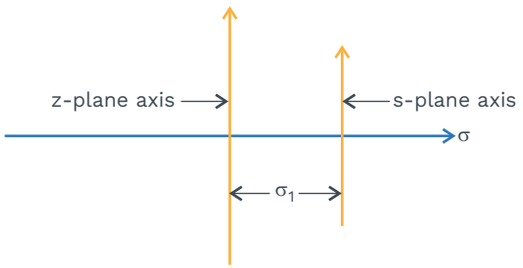


Figure-5.5

Previous Year's Questions

For given polynomial equation,
$s^3 + 5.5\ s^2 + 8.5\ s + 3 = 0$
The number of roots of the polynomial, which have the real parts strictly less than –1, is.
Answer: 2

- When the Routh criterion is satisfied by a new characteristic equation in the z-plane, all the roots of the original characteristic equation should be more negative than $-\sigma_1$.

Example-16.

A third order system has a characteristic equation given as,

$$s^3 + 7s^2 + 25s + 39 = 0$$

Check whether the roots are lying more negatively w.r.t. –1 or not.

Solution:

The characteristic equation is -

$$s^3 + 7s^2 + 25s + 39 = 0$$

Shift the origin to s = –1 by substitution,

$$s = z - 1$$

In the characteristic equation, then –

$$z^3 + 4z^2 + 14z + 20 = 0$$

Routh array is,

z^3	1	14
z^2	4	20
z^1	9	
z^0	20	

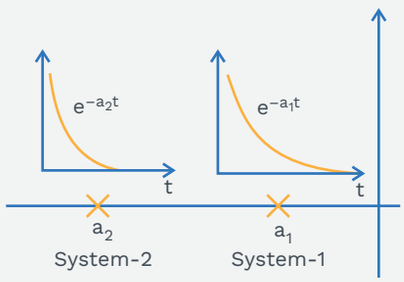
There is no sign change in first column therefore all the roots are lies in the left half of z-plane.

$\therefore$ All roots of original characteristics equation in the s-domain lie to left of s = –1.

Limitation of Routh Hurwitz criterion –

- Routh Hurwitz criterion is only applicable on the systems which are linear.
- It doesn't provide the accurate location of poles lying on the left and right side of the s-plane.
- The characteristic equation of a polynomial should have real and positive coefficient.
- It evaluates the stability of the system but doesn't allow an unstable system to get stabilized.
- Relative stability using Routh array is not feasible for higher order polynomials because it involves shifting of origin of the s-plane more negatively.

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- Both system-1 and system-2 are absolutely stable.
- System-2 is relatively more stable than system-1.

PRACTICE QUESTIONS

Q1 **The characteristic equation of a system is given as–**

$s^4 + 4s^3 + 13s^2 + 36s + K = 0$

Calculate the range of 'K' for which the system is said to be stable.

Sol. $36 > K > 0$

Q2 **A unity feedback control system has an OLTF which is given as,**

$$G(s) = \frac{K}{s(1+sT_1)(1+sT_2)}$$

Determine the value of K in term of T_1 and T_2 for the system to be stable.

Sol. $0 < K < \left(\frac{1}{T_1} + \frac{1}{T_2}\right)$

Q3 **Find the number of poles in the right half plane, left half plane and on imaginary axis respectively if transfer function is given as**

$$T(s) = \frac{s^2 + 7s + 10}{s^6 + 2s^5 - s^2 - 2}$$

Sol. 3, 3, 0

Q4 **A unity feedback system is characterized by the open loop transfer function –**

$$G(s) = \frac{K(s+13)}{s(s+3)(s+7)}$$

With the help of Routh criterion, calculate the range of 'K' for which the system is said to be stable.

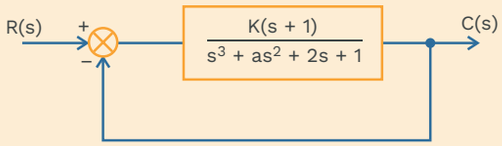
Sol. $0 < K < 7.5$

Q5 **Using Routh Hurwitz criterion determines the relation between 'K' and 'T'. So that unity feedback control signal whose transfer function is as –**

$$G(s) = \frac{K}{s[s(s+10)+T]}$$

Sol. **$0 < K < 10\ T$**

Q6 **A system oscillates with frequency ω, if it has poles at $s = \pm j\omega$ and no poles in the right half s-plane. Determines the values of K and a, so that the system shown below oscillates at a frequency of 2 rad/sec.**



Sol. **$K = 2, \quad a = 0.75$**

Q7 **A unity feedback system, having an open loop gain $G(s)\,H(s) = \frac{K(1-s)}{1+s}$ becomes stable when**

(a) $|K| > 1$ **(b) $|K| < 1$** **(c) $K > 1$** **(d) $K < -1$**

Sol. **Correct answer is (b).**

Q8 **For the given characteristic equation,**

$s^3 + 3s^2 + 2s + K = 0$

What will be the range of K when roots of the equation lies in left half of the s-plane?

(a) $0 < K < 6$ **(b)** $6 < K < 36$ **(c)** $0 < K < 16$ **(d)** $6 < K < 16$

Sol. **Correct answer is (a).**

Q9 **A unity feedback control system has a open loop transfer function given as,**

$$G(s) = \frac{K}{(s+2)(s+4)(s^2+6s+25)}$$

What values of K that will lead to sustained oscillations in the closed loop system. Also calculate their oscillation frequencies.

Sol. **K = 666.25, ω = 4.06 rad/sec**

Q10 **The LTI system characteristic equation in the negative unity feedback configuration is given by, $s^4 + 2s^2 + 3 = 0$. The number of poles located on jω axis is _____.**

Sol. **0**